

Technical Documentation:
Basic Logic Gates Implementation

COMPANY NAME : CODTECH IT SOLUTIONS

INTERN NAME : KAKARLAMUDI DIVYA

INTERN ID : CITS998

DOMAIN : VLSI DESIGN

TASK TITLE: BASIC LOGIC GATES

Project Overview:

This project implements the fundamental logic gates in Verilog HDL using Dataflow modeling. It includes a comprehensive testbench to verify the functionality of all gates across all possible binary input combinations.

Design Module

(basic_gates.v)The design module utilizes continuous assignment (assign statements) with bitwise operators to implement seven basic logic functions.

Port name	Direction	Type	Discription
a	Input	Wire	Primary Input Signal A
b	Input	Wire	Primary input signal B
y_and	Output	Wire	Output of 2-input AND

			gate ($(A \cdot B)$)
y_or	Output	Wire	Output of 2-input OR gate ($(A + B)$)
y_not	Output	Wire	Output of NOT gate (Inverts Input $(A \rightarrow \bar{A})$)
y_nand	Output	Wire	Output of 2-input NAND gate ($(\overline{A \cdot B})$)
y_nor	Output	Wire	Output of 2-input NOR gate ($(\overline{A + B})$)
y_xor	Output	Wire	Output of 2-input XOR gate ($(A \oplus B)$)
y_xnor	Output	Wire	Output of 2-input XNOR gate ($(\overline{A \oplus B})$)

2. Testbench Module:

(tb_basic_gates.v) The testbench validates the design by generating stimulus for all four binary combinations ($2^2 = 4$) with a time interval of 10ns between each test case.

Key Verification Features:

UUT Instantiation:

Maps testbench registers (tb_a, tb_b) to design inputs and wires to design outputs.

VCD Dumping:

Includes \$dumpfile and \$dumpvars to generate waveform data (dump.vcd) for graphical analysis.

Console Monitoring:

Uses \$monitor to display real-time signal changes in the simulation console.

3.Theoretical Truth Table

Based on the logic equations, here is the expected output behavior:

Time	Input A	Input B	AND	OR	NOT_A	NAND	NOR	XOR	XNOR
0ns	0	0	0	0	1	1	1	0	1
10ns	0	1	0	1	1	1	0	1	0
20ns	1	0	0	1	0	1	0	1	0
30ns	1	1	1	1	0	0	0	0	1

4.Source code :

Verilog source code:

```
//  
=====
```

// Company Name: CODTECH IT SOLUTIONS
// Intern Name: Kakarlamudi Divya
// Intern ID: CITS998
// Domain: VLSI

// Task: Basic Logic Gates (Verilog Implementation)

//

=====

``timescale 1ns/1ps`

`module basic_gates (`

`input wire a, // Input Signal A`

`input wire b, // Input Signal B`

`output wire y_and, // AND Gate Output`

`output wire y_or, // OR Gate Output`

`output wire y_not, // NOT Gate Output (Inverts 'a')`

`output wire y_nand, // NAND Gate Output`

`output wire y_nor, // NOR Gate Output`

`output wire y_xor, // XOR Gate Output`

`output wire y_xnor // XNOR Gate Output`

`);`

`// Continuous assignment using bitwise operators (Dataflow Modeling)`

`assign y_and = a & b; // Logic AND`

`assign y_or = a | b; // Logic OR`

`assign y_not = ~a; // Logic NOT`

`assign y_nand = ~(a & b); // Logic NAND`

`assign y_nor = ~(a | b); // Logic NOR`

`assign y_xor = a ^ b; // Logic XOR`

`assign y_xnor = ~(a ^ b); // Logic XNOR`

`endmodule`

Testbench code :

```
//
=====
=====
// Company Name: CODTECH IT SOLUTIONS
// Intern Name: Kakarlamudi Divya
// Intern ID: CITS998
// Domain: VLSI
// Task: Basic Logic Gates Testbench
//
=====
=====
```

```
`timescale 1ns / 1ps
```

```
module tb_basic_gates;
```

```
    // Inputs declaration (reg type for testbenches)
```

```
    reg tb_a;
```

```
    reg tb_b;
```

```
    // Outputs declaration (wire type to monitor UUT)
```

```
    wire tb_and, tb_or, tb_not, tb_nand, tb_nor, tb_xor, tb_xnor;
```

```
    // Instantiate the Unit Under Test (UUT)
```

```
    basic_gates uut (
```

```
        .a(tb_a), .b(tb_b),
```

```
        .y_and(tb_and), .y_or(tb_or), .y_not(tb_not),
```

```
        .y_nand(tb_nand), .y_nor(tb_nor), .y_xor(tb_xor),
```

```
        .y_xnor(tb_xnor)
```

```
    );
```

```
    // Stimulus block to test all 4 binary combinations
```

```
    initial begin
```

```

    $dumpfile("dump.vcd");
    $dumpvars;
end
    initial begin
        // Display format for simulation console
        $monitor("Time=%0td | Input A=%b B=%b | AND=%b
OR=%b NOT_A=%b NAND=%b NOR=%b XOR=%b
XNOR=%b",
            $time, tb_a, tb_b, tb_and, tb_or, tb_not, tb_nand,
            tb_nor, tb_xor, tb_xnor);

        // Case 1: Binary 00
        tb_a = 0; tb_b = 0; #10;

        // Case 2: Binary 01
        tb_a = 0; tb_b = 1; #10;

        // Case 3: Binary 10
        tb_a = 1; tb_b = 0; #10;

        // Case 4: Binary 11
        tb_a = 1; tb_b = 1; #10;

        // Finish simulation
        $finish;
    end

endmodule

```


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EDA playground

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• Languages & Libraries

Testbench • Design

SystemVerilog • UVM / OVM

Other Libraries

None

None

None

None

Tools & Simulators

None

None

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Testbench

1 // Company name: CODESIGN IT SOLUTIONS

2 // Project name: testbench_top

3 // Author: John Doe

4 // Date: 2023-10-27

5 // Description: Basic logic gates

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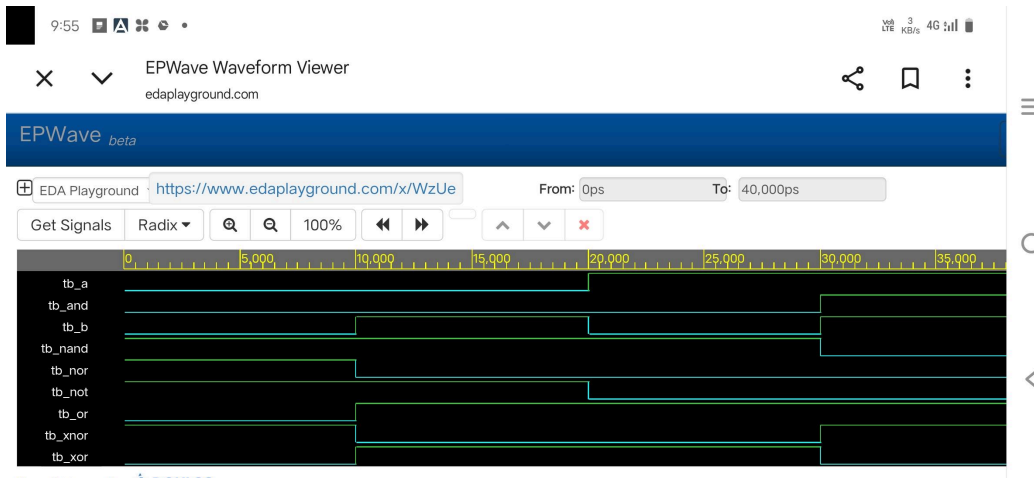
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6. Timing waveforms (epwave window)



7.Conclusion:

The RTL code successfully implements the boolean logic equations for all basic gates. The testbench accurately verifies the design functionality against standard truth tables, making it ready for synthesis and implementation.